

FOODBUK

IS Project



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Chapter One: Introduction.

Background

The background of this project centers on the dining facilities available to students on campus. At many educational institutions, including our own, students face the challenge of obtaining breakfast in the morning due to the rush and limited serving hours at the cafeteria. This issue often leads to compromises in meal choices, delays, and unnecessary stress. Addressing this problem is essential to enhance the overall campus experience and promote a healthier lifestyle for students.

...I have missed breakfast because I woke up late on several occasions or missing dinner because I was getting late to get back to school... Also the girl's hostels are quite far so sometimes someone is about lazy to go for meals..Then when you go you find out that what you wanted to eat isn't even available

-Yvonne

Problem Statement

The primary problem this project aims to address is the difficulty faced by students in obtaining breakfast from the cafeteria in the morning. The existing system does not provide an efficient and convenient way for students to pre-book their breakfast choices, resulting in long queues and limited meal options. This situation negatively impacts the students' productivity, time management, and overall well-being. Refer to the anecdotal evidence in figure 1-0.

Figure 1- 1 Testimonials from a student that boards on campus

Justification of the Problem (Significance)

Ensuring that students have access to a healthy and timely breakfast is crucial for their academic performance and general health. Numerous studies emphasize the significance of a nutritious breakfast in enhancing cognitive function, concentration, and academic achievements. By addressing the breakfast acquisition issue, this project contributes to the overall well-being and academic success of students on campus.

Objectives

The objectives of this project are as follows:

1. **Streamlining Breakfast Booking:** Develop a system that allows students to pre-book their breakfast choices, ensuring they get their preferred meals without queuing.
2. **Enhancing Efficiency:** Implement a mechanism that optimizes the cafeteria's breakfast serving process, reducing waiting times and improving service.
3. **Improving Time Management:** Enable students to plan their morning routines more effectively by reducing the time spent waiting for breakfast.
4. **Enhancing User Experience and feedback:**
 - i. **Engagement Forum System:** Implement an engagement forum system for food items ordered by students, enabling students to provide feedback and share their experiences with other students.
 - ii. **Complaints Management:** Develop a platform to address and manage complaints, allowing students to report issues related to food quality, service, or any other concerns.
5. **Remote Menu Viewing:** Allow users to view the available breakfast options remotely, enabling them to make informed choices before arriving at the cafeteria.

Research Questions

This project aims to answer the following research questions:

1. How can we design an effective pre-booking system for breakfast at the campus cafeteria?
2. What technological solutions can be employed to optimize the breakfast serving process and minimize waiting times?
3. How can we integrate nutritional information into the pre-booking system to promote healthier eating habits among students?

Assumptions and Limitations

Assumptions

- Students are willing to adopt a new breakfast pre-booking system for its convenience and time-saving benefits.
- The cafeteria staff will adapt to the new technological changes and work collaboratively to implement and utilize the pre-booking system effectively.
- Sufficient technological infrastructure is available to support the implementation of the pre-booking system.
- Students will login with their school email credentials provided

Limitations

- Financial constraints may limit the extent and complexity of the technological solutions that can be implemented.
- Resistance to change among some students and cafeteria staff may hinder the successful adoption and utilization of the pre-booking system.
- The effectiveness of the pre-booking system in reducing waiting times and promoting healthier eating habits will need to be evaluated and may vary based on individual preferences and behaviors.
- Regular Updates: The system will require consistent updates and maintenance by designated personnel to ensure its functionality, relevance, and alignment with evolving user needs and technological advancements.

FoodBuk

Explanation of Different Users

Client

Here the client logs into their account using their school email and gets verified, then access is granted. These types of users can only view the menu, and book/order food items and pay for them, refer to image in figure 1-2 for the view,.

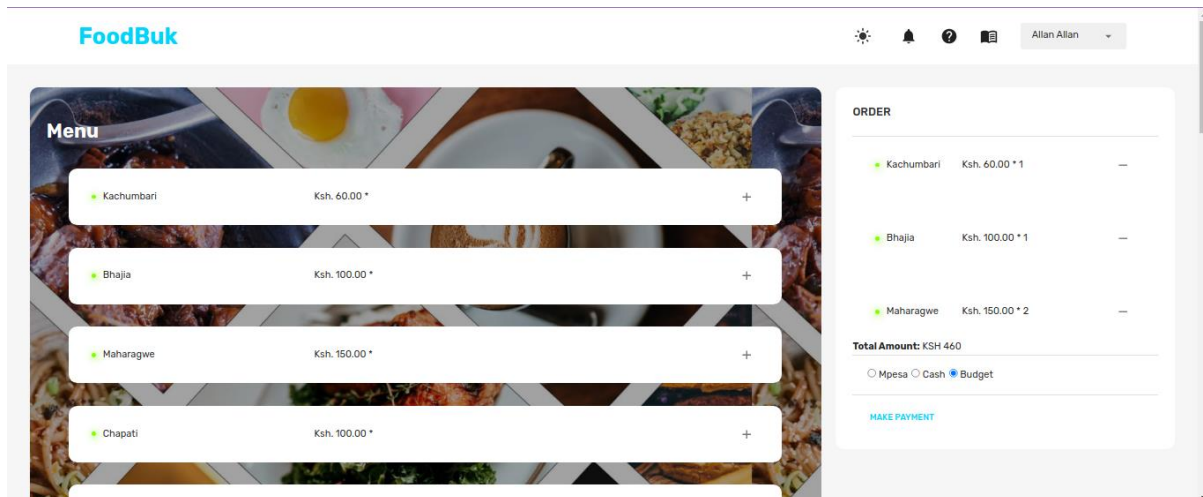


Figure 1- 2 Menu View of the food available and an order list widget to the right

Also engage with other students by posting the food they had and give it a rating, refer to image in figure 1-3, they can also complain and look at the history of their past purchases.

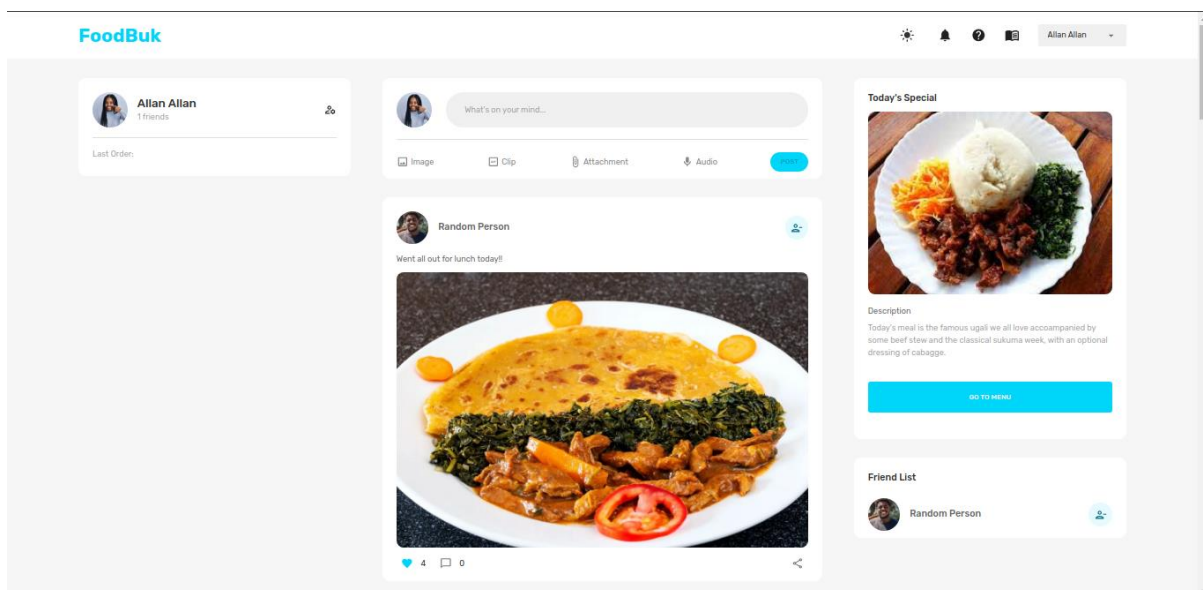


Figure 1- 3 Student engagement section of the app.

Admin

Here users can access the menu and modify the available meals and those that are not available for real time update of the database for more accurate booking of meals, they also receive meals orders and set them aside for pick up by the student, they also accept payment through cash if student selected the cash option.

Chapter 2: Literature Review

Introduction

In this chapter we explore the existing literature on online food ordering systems that share or resemble the same themes as that of this project. The aim is to understand the current state of these systems, their features, benefits and challenges, and how they have been implemented and used in different and unique contexts.

Global Literature Review

Globally, online food ordering system has become increasingly popular as a result of the convenience it offers along side the efficiency it provides (GloriaFood, 2023) (UpMenu, 2023) (Jotform, 2023) (Nelson, 2022).

Many companies such as GloriaFood and UpMenu offer comprehensive solutions that allow restaurants to accept orders online, integrate them with payment gateways, and manage their interfaces, rich functionalities and excellent customer support (Nelson, 2022).

However even then users have pointed out some areas for improvement such as more flexible integration options, for reference check out the one of the reviews in figure 2-1.

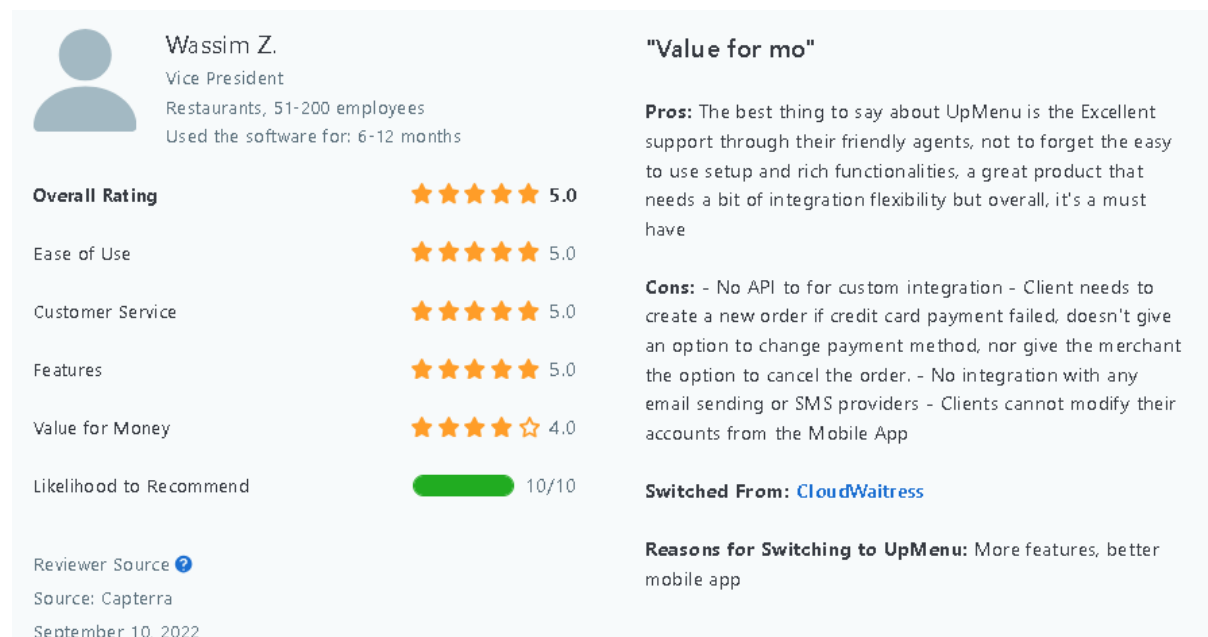


Figure 2- 1 A review of the food ordering systems by a customer Wassim Z.

Regional Literature Review

Regionally, the embracing of online food ordering systems varies, where in some area, these systems are widely use and have significantly changed the food service industry. For example, with in the country fast food chain restaurant being most common to use them, especially with the integration of delivery services such as uber eats and bolt among many other service providers. This has revolutionized the food industry tremendously

According to statista.com, the revenue in the Online Food Delivery market in Kenya is projected to reach KES US\$374.90m in 2023, showing an annual growth rate (CAGR 2023-2027) of 19.33%, resulting in a projected market volume of KES US\$760.30m by 2027. This growth is driven by the growing middle class and increasing smartphone penetration (Statista, 2023).

Local Literature Review

With in our locality Nairobi, the online food ordering landscape is vibrant and diverse with several key players offering unique services, from companies such as Glovo, Yum Deliveries and Jumia Food.

These platforms have significantly impacted the food industry in Nairobi, providing residents with a convenient way of accessing a wide variety of food options from the comfort of their homes.

Summary

The literature review reveals that online food ordering systems have become an integral part of the food service industry at global, regional, and local levels. These systems, represented by companies like GloriaFood and UpMenu that some what resemble the FoodBuk objectives and others Like UberEats and Glovo that share concepts such as a user getting food they order and also being able to know availability of meals based of the closing times.

Chapter 3: Methodology

Introduction

This chapter outlines the methodology used in the development of FoodBuk.

The methodology includes the approach, data gathering methods, field intervention, sampling, system analysis, and test plan.

Locality and Beneficiary of the Project

The project is localized within the university campus, mostly targeting students who live on campus and rely on the school cafeteria for their meals. The primary beneficiaries of the project are these students, who will be able to view the menu, see available meals, and reserve meals remotely from their rooms.

Sample

The target group for this project is university students living on camps. These students are chosen because they are the primary users of the school cafeteria and face challenges in accessing meals due to time constraints and limited serving hours.

Sampling Method

The sampling method used for this project was random sampling. This method was chosen to ensure a fair representation of the student population and to avoid bias in the data collected.

Data Collection Methods

Data was collected through anecdotes.

A message was sent to a couple of individuals asking them to share their experiences with the current dining system and how FoodBuk could improve their daily experience.

The message served as an informal interview guide, prompting respondents to provide anecdotal evidence to support the claims made in the project.

Data Analysis Methods

The data collected was analyzed by using qualitative content analysis.

This involved reading through the responses, identifying common themes and patterns, and interpreting the data in the context of the project objectives.

System Analysis and Design Methods

The system analysis and design for FoodBuk followed a user-centered approach.

This involved understanding the needs and preferences of the users (students), defining the system requirements based on these needs, and designing the system interface and functionality to meet these requirements.

The design process also considered the technical constraints and requirements of the cafeteria's operations.

Testing Plan For The System

The testing plan for FoodBuk includes usability testing to ensure the system is easy to use and meets the needs of the users.

It also includes performance testing to ensure the system can handle the expected number of users and transactions.

The testing results will be used to identify and fix any issues before the system is fully implemented.

In conclusion, the methodology used in this project ensures a thorough and systematic approach to the development of FoodBuk, ensuring it effectively addresses the needs of the students and enhances their dining experience.

Chapter 4: Systems Analysis and System Design

Introduction

In this segment, we would look at the modeling and design of the systems in terms of the data flow and storage

Functional Requirements

The functional requirements of the system being developed include as follows;

1. Users must be able to log in to the system
2. Users must be able to post on the engagement section.
3. Users must be able to view, comment and like other individuals posts
4. Users must be able to add friends
5. Users must be able to navigate to menu and view available food items and non-available ones
6. Users must be able to add food items to order list and preview the order list before placing the final order and also remove items.
7. Users must be able to see the orders they have place (both current and history)
8. Users must be able to send complaints

Non- Functional Requirements

1. Performance Requirements- should be speedy and have a fast response, and update live.
2. Scalability Requirements- the server should handle all students in the school.
3. Security Requirements- all user passwords should be encrypted.
4. Usability Requirements- all system functions are accessible from the homepage of the app.
5. Reliability Requirements should be able to function at all times.
6. Availability Requirements- the system should be available 24/7.
7. Maintainability Requirements- any bugs reported should be fixed within 48 hours.
8. Portability Requirements- the software should be platform-independent.

Use Case Diagrams

This is a representation of what a user can do in the system using a use case diagram in figure 4-1.

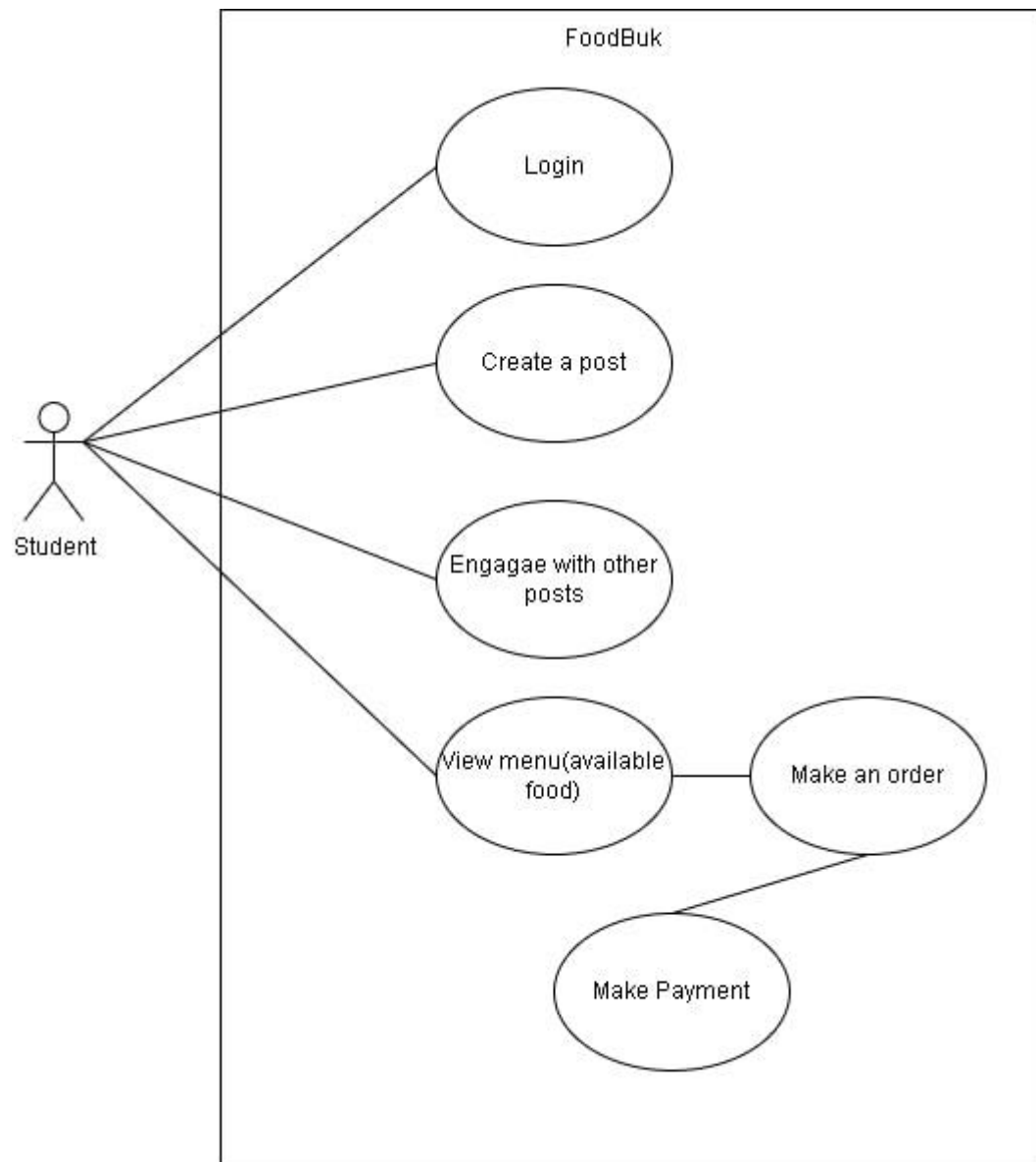


Figure 4- 1 A use case diagram of the FoodBuk Web app

Context Diagrams

This a representation of process and outputs gotten for users using a context diagram.

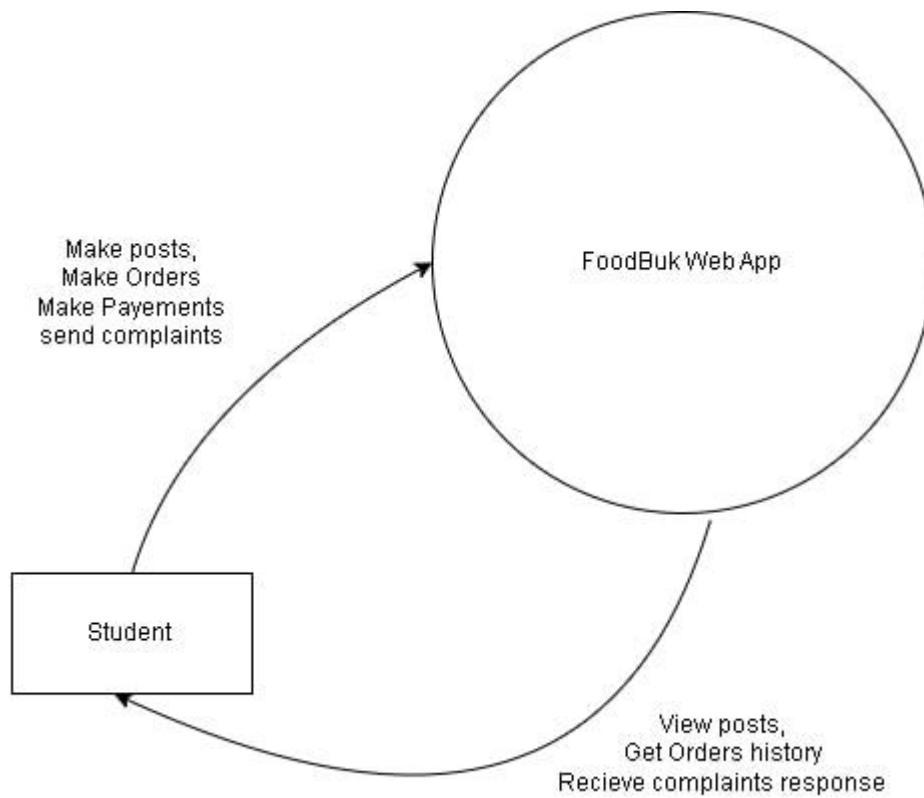


Figure 4- 2 A use case Diagram of what the user can do within the system

Data Flow Diagrams

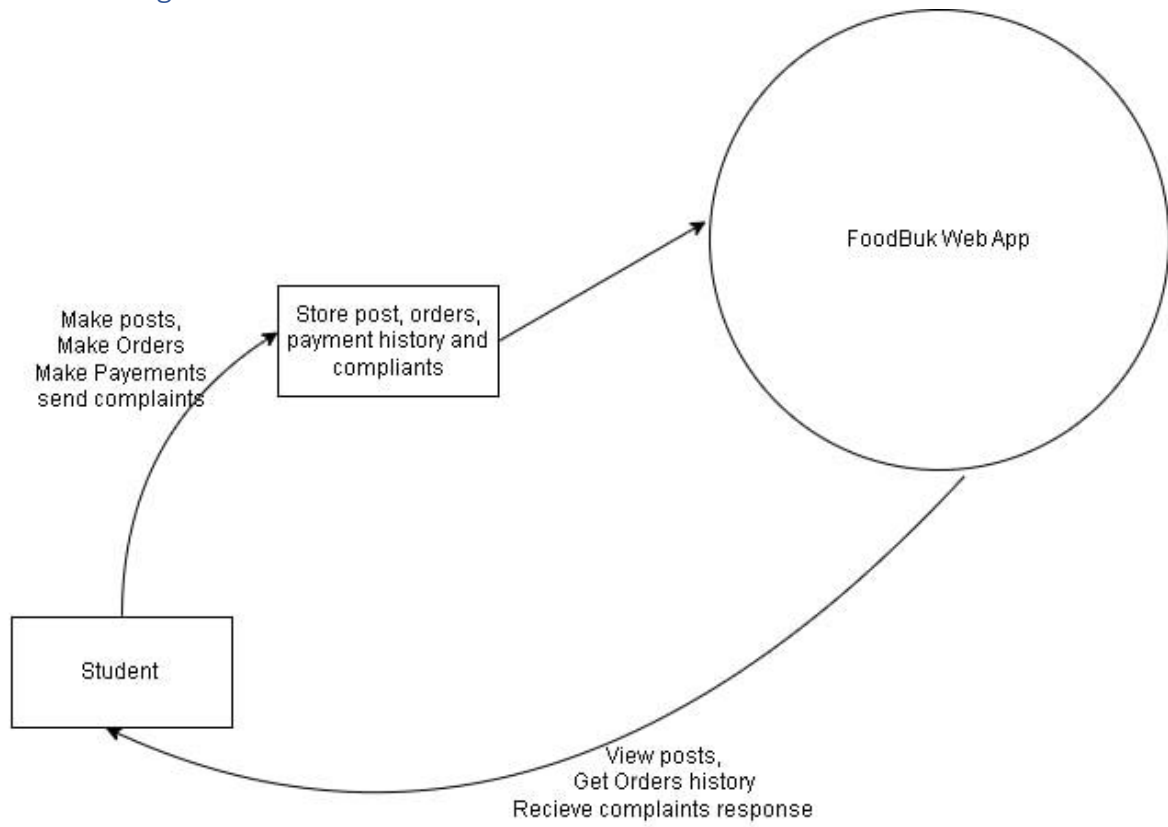


Figure 4- 3 A Data Flow Diagram of the system and showing what is stored in the system.

Chapter 5: Implementation (Prototype)

Prototype Design and Implementation

The following is the structure of the database prototype and what it should resemble

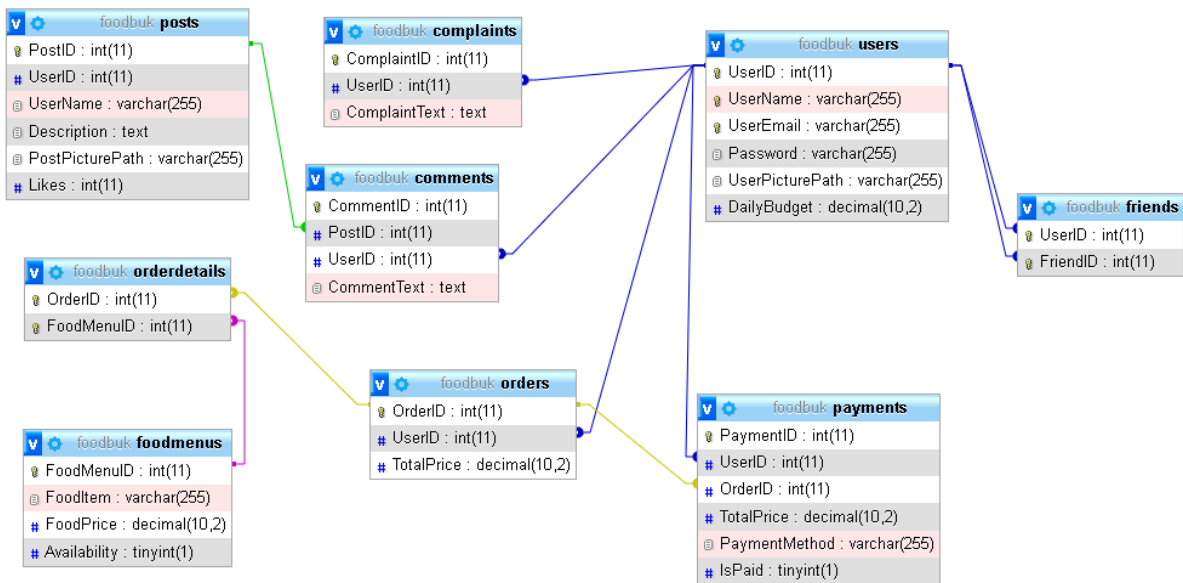


Figure 5- 1 Visual representation of what the database would look like in a sql 3NF database

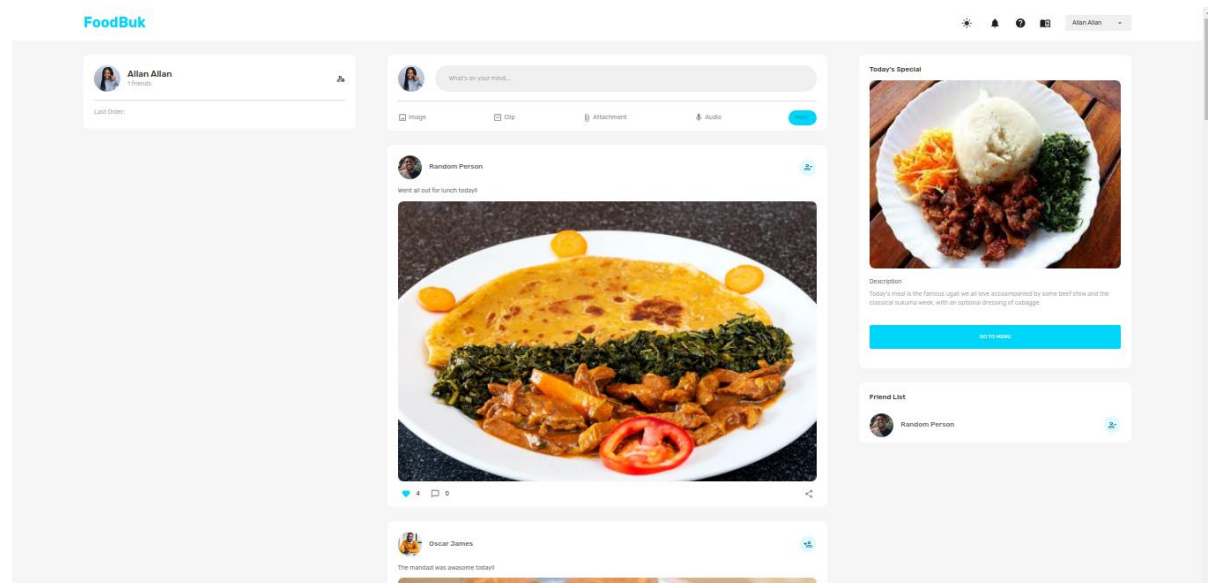


Figure 5- 2 Visual representation of the Engagment forum

Describe your complaint

Complaint

SUBMIT

Figure 5- 3 Prototype of the complaint page

FoodBuk

Menu

Item	Price	Quantity
Kachumbani	Ksh. 50.00 *	1
Bhalla	Ksh. 100.00 *	1
Muhang'ee	Ksh. 150.00 *	1
Chapati	Ksh. 100.00 *	1
Sukuma Wiki with Chapati	Ksh. 180.00 *	1
Muruk	Ksh. 70.00 *	1
Nyama Choma	Ksh. 450.00 *	1
Gitoni	Ksh. 500.00 *	1
Mohariro	Ksh. 70.00 *	1
Mandazi	Ksh. 30.00 *	1

ORDER

Item	Price	Quantity
Kachumbani	Ksh. 50.00 *	1
Bhalla	Ksh. 100.00 *	1
Muhang'ee	Ksh. 150.00 *	2

Total Amount: Ksh. 450

☐ Mpasa ☐ Cash ☒ Budget

[View Payment](#)

Figure 5- 4 Possible presentation of the menu page together with the order list section

Conclusion and Recommendations

Conclusion

Foodbuk is a project on-campus web app that will help students who live on campus to book food from the school cafeteria and also share their experiences while there. The program has achieved its objectives of providing convenience, variety, and social interaction for the students. Foodbuk will also benefited the cafeteria by customer satisfaction, and feedback. The program has demonstrated the potential of online food delivery in the Kenyan market, which is projected to grow significantly in the next few years.

Recommendation

Based on the findings and analysis of the program, the following recommendations are suggested for future improvement and expansion:

Enhance the user interface and functionality of the Foodbuk app and website to make them more user-friendly, attractive, and responsive. For example, adding features such as ratings, reviews, filters, favorites, and notifications.

Expand the menu options and catering services of the cafeteria to offer more diversity and quality of food for the students. For example, introducing new cuisines, dishes, and dietary preferences, as well as catering for events and occasions.

Partner with other food vendors and delivery services on and off campus to increase the availability and accessibility of food for the students. For example, collaborating with local restaurants, cafes, and supermarkets, as well as online platforms such as Jumia Food and Uber Eats.

Conduct regular surveys and feedback sessions with the students and the cafeteria staff to monitor the performance and satisfaction of the program. For example, using online forms, polls, and focus groups to collect data and opinions.

Explore the possibility of scaling up the program to other campuses and institutions in Kenya and beyond. For example, conducting market research, feasibility studies, and pilot tests to assess the demand and viability of the program in different contexts and locations.

References

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